MP Online • Internship Project • 2025-26

Expense Tracker with Smart Insighter

User Guide Document

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MP Online Internship Project

1. Introduction:

The Expense Tracker Smart Insighter is a web-based application developed to help users efficiently manage their financial expenses. The system provides an intuitive interface for recording expenses and includes a smart categorization feature to automatically classify expenses.

2. Purpose:

The purpose of this application is to:

- Provide a platform for tracking daily expenses
- Automate expense categorization
- Improve financial awareness
- Assist users in analyzing spending behavior

3. Scope:

This system is designed for individual users who want to:

- Record and manage expenses
- Automatically categorize transactions
- View and organize financial data

The application currently focuses on core expense tracking and basic analytics.

4. System Overview:

The system follows a client-server architecture:

- Frontend: User interface built using React
- Backend: API services built using FastAPI
- Database: PostgreSQL for storing user and expense data

The system processes user input, categorizes expenses, and stores structured data for retrieval.

5. System Requirements:

5.1 Hardware Requirements:

- Minimum 4GB RAM
- Standard personal computer or laptop

5.2 Software Requirements:

- Web Browser (Chrome, Edge, Firefox)
- Node.js (for frontend)
- Python (for backend)
- PostgreSQL Database

6. User Roles:

Role	Description
User	Can register, login, and manage expenses

7. Features and Functionalities:

7.1 User Registration:

- Allows new users to create an account using name, email, and password

7.2 User Login:

- Authenticates users using registered credentials

7.3 Add Expense:

Users can enter:

- Amount
- Description
- Date

The system automatically assigns a category

7.4 View Expenses:

- Displays list of all expenses

Includes:

- Amount
- Description
- Category
- Date

7.5 Delete Expense:

- Users can remove unwanted expense records

7.6 Smart Categorization:

- Automatically classifies expenses based on keywords
- Improves usability by reducing manual input

8. Operating Instructions:

Step 1: Start Backend Server:

- uvicorn main:app --reload

Step 2: Start Frontend Application:

- npm install

  npm run dev

Step 3: Access Application:

- Open browser and navigate to:

  http://localhost:5173

Step 4: Using the System:

- Register a new account
- Login using credentials
- Add expenses

- View categorized data
- Delete entries if required

9. System Workflow:

1. User logs into the system
2. User adds an expense
3. Backend validates input
4. Categorization logic processes description
5. Data is stored in the database
6. User retrieves and views stored expenses

10. Error Handling:

- Invalid input is rejected using validation
- Incorrect login credentials are handled with error messages
- Backend errors are logged for debugging

11. Limitations:

- Basic authentication (no JWT implementation)
- Rule-based categorization (limited accuracy)
- No advanced filtering or pagination
- Limited scalability in current version

12. Future Enhancements:

- Implement JWT-based authentication
- Integrate machine learning for categorization
- Add graphical dashboards and analytics
- Improve performance and scalability
- Enhance user interface

13. Conclusion:

The Expense Tracker Smart Insighter provides a structured and efficient solution for managing expenses. It combines core expense tracking with intelligent categorization, making it a practical tool for improving financial awareness. Future improvements can further enhance its scalability and intelligence.